Arista Access Node MAR-ACC-30: BGP Route Churn, OSPF Adjacency Loss, and ISIS LSP Flooding Failure SOP

Node: MAR-ACC-30 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_5506

Description

This Standard Operating Procedure (SOP) addresses the critical failure domain of Routing Protocol Flap on the Arista Access Node MAR-ACC-30 at the Marathahalli Core site. The failure occurs due to the instability in the BGP route churn, OSPF adjacency loss, and ISIS LSP flooding, which are inherent properties of the routing protocols employed in this specific role and vendor equipment. The BGP route churn is caused by the frequent updates in the routing tables, leading to an increase in the number of routing updates and advertisements. The OSPF adjacency loss is triggered by the instability in the OSPF neighbor relationships, resulting in the loss of routing information and connectivity. The ISIS LSP flooding is caused by the excessive flooding of link state advertisements (LSAs), leading to a significant increase in the control plane traffic and instability. The root cause of these issues is often attributed to the misconfiguration, hardware failures, or software bugs in the Arista Access Node MAR-ACC-30.

Pre-requisite Checks

- Verify that the remote hands are dispatched to the Marathahalli Core site to physically inspect the equipment.

- Check the status of the redundant links to ensure that they are functioning properly.

- Verify that the switch is running the latest software version and that there are no pending software updates.

Potential Root Causes

Cause	Probability
Misconfiguration of BGP, OSPF, or ISIS settings	High
Hardware failure in the switch or network interface	Medium
Software bugs or glitches in the Arista Access Node MAR-ACC-30	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp summary, Expected Output: BGP summary information, Description: This command displays the BGP summary information, including the number of connections, updates, and advertisements. It helps to identify any issues with the BGP routing protocol.

Command: show ospf neighbor, Expected Output: OSPF neighbor information, Description: This command displays the OSPF neighbor information, including the neighbor IDs, states, and adjacency status.
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It helps to identify any issues with the OSPF routing protocol.

Command: show isis database, Expected Output: ISIS database information, Description: This command displays the ISIS database information, including the LSPs, LSAs, and adjacency status. It helps to identify any issues with the ISIS routing protocol.
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Resolution Steps

Step 1: Step: Verify and correct BGP, OSPF, or ISIS settings, Command: show run | i bgp ospf isis, Description: Verify the BGP, OSPF, or ISIS settings and correct any misconfigurations. This step helps to resolve the issues with the routing protocols.

Step 2: Step: Replace faulty hardware, Command: no switchport, Description: Replace any faulty hardware, such as the switch or network interface, to resolve the issues with the hardware.

Step 3: Step: Update software or firmware, Command: request system software save, Description: Update the software or firmware to the latest version to resolve any software bugs or glitches.

Post-Check Verification

```
Command: show bgp summary, Expected Output: BGP summary information, Description: Verify that the BGP routing protocol is stable and functioning properly.

Command: show ospf neighbor, Expected Output: OSPF neighbor information, Description: Verify that the OSPF routing protocol is stable and functioning properly.

Command: show isis database, Expected Output: ISIS database information, Description: Verify that the ISIS routing protocol is stable and functioning properly.
```

Roll-back Procedures

- Step: Restore previous configuration, Command: copy run start, Description: Restore the previous configuration to resolve any issues caused by the new configuration.

- Step: Revert software or firmware, Command: request system software revert, Description: Revert the software or firmware to the previous version to resolve any issues caused by the new software or firmware.

- Step: Replace hardware if necessary, Command: no switchport, Description: Replace any faulty hardware, such as the switch or network interface, if necessary to resolve any issues.